SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION **COURSE CODE:** DSA0204
COURSE OUTCOME COVERED

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CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

Viola-Jones: fast but less accurate (78%) YOLO: moderate speed, good accuracy (92%) Deep Learning: high accuracy (95%) but slow

Answer:

Viola-Jones offers the highest speed but its 78% accuracy is too low for reliable surveillance and risks missed detections. Deep Learning gives the best accuracy (95%) but its high latency violates the strict real-time constraint. YOLO strikes the best balance: 92% accuracy at moderate latency satisfies the real-time requirement while keeping accuracy high enough for practical use.

Conclusion: YOLO is the optimal choice under strict latency constraints while maintaining acceptable accuracy, since it is the only option that satisfies both conditions simultaneously; Viola-Jones sacrifices too much accuracy and Deep Learning breaches the latency budget.

2. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Answer:

Average accuracy: Viola-Jones = (80+60+55)/3 = 65%, YOLO = (90+75+70)/3 = 78.3%, Deep Learning = (95+85+80)/3 = 86.7%.

Deep Learning has both the highest average accuracy and the smallest degradation from normal to low-light conditions (a 15-point drop, versus 20 for YOLO and 25 for Viola-Jones), making it the most robust technique overall across varying environmental conditions.

Trade-off: this robustness comes at the cost of higher computation, so Deep Learning is recommended when robustness is prioritized over speed; YOLO is a reasonable compromise when moderate compute is available.

3. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

Method A: stable vectors but misses small motions Method B: detects fine motion but noisy vectors

Answer:

For motion tracking in dynamic scenes that require capturing fine, rapid movement, Method B is more suitable because it detects subtle motion that Method A would miss — provided its noise is controlled with a smoothing or Kalman filter.

Method A is preferable in applications where reliability and stability matter more than fine detail, such as coarse tripwire-style surveillance.

Conclusion: for general dynamic-scene tracking, Method B combined with a noise-reduction post-processing stage offers the best overall trade-off.

4. Background Modeling Strategy (GMM)

A GMM-based system shows:

High false positives in dynamic background Missed detections in static background

Answer:

The GMM system struggles at both ends: it flags moving background elements (trees, water, shadows) as foreground in dynamic scenes, while slow-moving or stationary objects blend into the background model and go undetected in static scenes.

Conclusion: GMM alone is not sufficient. Recommended improvements include adaptive learning-rate control, shadow removal, morphological post-processing to suppress dynamic-background noise, and hybridizing GMM with optical flow or a deep-learning-based background subtraction method to catch static/slow-moving foreground objects.

5. Motion Estimation vs Computational Cost

Two methods:

Method A: low error (5%), high computation Method B: moderate error (10%), low computation

Answer:

Method A's low error makes it suitable for offline or accuracy-critical applications (e.g., video post-production, medical imaging) where computation time is not a limiting factor.

Method B's low computational cost makes it better suited for real-time applications (e.g., live tracking, robotics) where a moderate 10% error is an acceptable trade-off for speed.

Conclusion: choose Method A for offline/high-precision tasks and Method B for real-time deployments.

6. Structure from Motion Consistency

Two datasets produce:

Dataset A: accurate depth but requires high-quality images Dataset B: inconsistent depth but works on low-quality images

Answer:

Real-world deployments (mobile, outdoor, uncontrolled lighting) frequently involve variable image quality, motion blur, and compression artifacts, so Dataset B's tolerance for low-quality input makes its setup more practical for general deployment, provided post-processing or error-correction is applied to reduce inconsistency.

Where image capture can be controlled and high precision is essential (e.g., engineering-grade 3D reconstruction), Dataset A's setup is preferable.

Conclusion: for general real-world use, Dataset B is more practical; for controlled high-precision use cases, Dataset A is superior.

7. Stereo Vision vs Motion Estimation

Depth estimation results:

Stereo Vision: accurate but requires dual cameras Motion Estimation: less accurate but works with single camera

Answer:

Under hardware constraints where only a single camera is available (e.g., typical smartphones, low-cost embedded systems), Motion Estimation is the only feasible option despite its lower accuracy.

When dual-camera hardware is available and precision is critical (e.g., robotic manipulation, AR depth sensing), Stereo Vision is the preferred choice.

Conclusion: the decision is dictated primarily by available hardware — single camera favors Motion Estimation; dual camera favors Stereo Vision.

8. Model Generalization vs Overfitting

Performance:

Model	Training	Testing

A	98%	70%
B	90%	88%

Answer:

Generalization gap: Model A = 98% − 70% = 28 points (severe overfitting). Model B = 90% − 88% = 2 points (good generalization).

Although Model A has higher training accuracy, its large train-test gap shows it has memorized the training data and will likely perform poorly on unseen real-world inputs.

Conclusion: Model B is more reliable for deployment because it generalizes well, despite its slightly lower training accuracy.

9. YOLO Performance across Scales

Detection results:

Large objects: 95% Medium objects: 85% Small objects: 60%

Answer:

There is a steep accuracy decline as object size decreases (95% → 85% → 60%), a well-known limitation of YOLO caused by the loss of fine spatial detail during repeated downsampling in its feature maps.

Conclusion: YOLO alone is not sufficient for multi-scale detection tasks, particularly small-object detection. Recommended improvements: Feature Pyramid Networks (FPN), higher input resolution, multi-scale training/augmentation, finer anchor box tuning, and using small-object-optimized variants (e.g., YOLO with an added P2 detection head).

10. Motion Model Selection for Animation

Two models:

Model A: smooth but physically inaccurate Model B: realistic but computationally intensive

Answer:

Model A's smoothness with low computational cost makes it appropriate for real-time gameplay, where fluid interactive motion matters more than strict physical accuracy.

Model B's realism makes it suited to cinematic, pre-rendered sequences where visual fidelity is paramount and rendering time is not constrained.

Conclusion: use Model A for real-time interactive systems and Model B for offline cinematic rendering — a hybrid engine can apply each model contextually.

11. Detection Model Comparison under Resource Constraints

Performance:

Model	Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

Answer:

Model A is fastest (40 ms) but has the lowest accuracy; Model B has the highest accuracy but its 90 ms latency is too high for many real-time systems; Model C offers a strong middle ground of 90% accuracy at 60 ms.

Conclusion: Model C is optimal for real-time systems with moderate accuracy requirements, as it balances speed and accuracy better than the other two options.

12. Robustness vs Speed Trade-off

Performance:

Method	Accuracy	Robustness	Speed
A	88%	High	Slow
B	85%	Moderate	Fast
C	90%	Low	Moderate

Answer:

No single method dominates on all three metrics: Method A is best on robustness, Method C is best on accuracy, and Method B is best on speed.

For a genuinely balanced system, Method A is recommended when correctness under varying/adverse conditions is the priority (its high robustness outweighs its slower speed and slightly lower accuracy than C); Method B is preferable when raw speed is the priority; Method C when peak accuracy under stable conditions matters most.

Conclusion: given the requirement is a 'balanced' system, Method A offers the best overall combination of strong accuracy and high robustness, accepting a speed trade-off.

13. Optical Flow Stability Analysis

Results:

Method	Accuracy	Noise Sensitivity
A	High	High
B	Moderate	Low

Answer:

In noisy environments, Method A's high noise sensitivity would generate erratic or false motion vectors despite its higher raw accuracy under clean conditions.

Method B's low noise sensitivity keeps its output stable even though its baseline accuracy is only moderate.

Conclusion: Method B is more suitable for real-time tracking in noisy environments, since stability under noise outweighs a modest accuracy advantage that is unreliable in practice.

14. Motion Estimation Accuracy vs Efficiency

Results:

Method	Error	Computation
A	4%	High
B	9%	Low
C	6%	Moderate

Answer:

Method A is most accurate but too computationally expensive for embedded hardware; Method B is efficient but its error is more than double Method A's; Method C offers a middle-ground 6% error at moderate computation.

Conclusion: Method C is the best approach for embedded systems, since it keeps computation within typical embedded budgets while maintaining a reasonably low error rate.

15. GMM Performance in Different Conditions

Results:

Scenario	Accuracy
Static	92%
Dynamic	68%

Answer:

The 24-percentage-point drop from static (92%) to dynamic (68%) scenes shows GMM is reliable only in relatively static environments and is not dependable across all environments.

Conclusion: GMM alone is not reliable for all environments. Suggested improvements: adaptive GMM with dynamic learning-rate tuning, shadow suppression, or hybridizing with deep-learning-based background subtraction (e.g., FgSegNet) to better handle dynamic scenes.

16. Object Detection under Scale Variation

Results:

Object Size	Accuracy
Large	94%
Medium	80%
Small	55%

Answer:

Accuracy degrades sharply as object size decreases, indicating the detector loses effective resolution/feature detail for small objects — a common limitation tied to anchor size and downsampling stride.

Conclusion: the system has a clear small-object detection limitation. Improvements: multi-scale feature pyramids, higher input resolution or image tiling, and data augmentation that oversamples small-object examples.

17. Stereo vs SfM Accuracy Comparison

Results:

Method	Accuracy	Hardware
Stereo	High	Dual camera
SfM	Moderate	Single camera

Answer:

Under hardware limitations where only a single camera is available, Structure from Motion (SfM) is the only practical option despite its moderate accuracy.

When dual-camera hardware is available and highest accuracy is required, Stereo Vision is preferable.

Conclusion: the choice is primarily hardware-driven — SfM for single-camera systems, Stereo for dual-camera systems needing higher precision.

18. Detection Model Generalization

Results:

Model	Training	Testing
A	97%	75%
B	90%	88%

Answer:

Generalization gap: Model A = 97% − 75% = 22 points; Model B = 90% − 88% = 2 points.

Conclusion: Model B generalizes far better to unseen data and is the better choice for deployment; Model A's large gap indicates overfitting to the training set.

19. Motion Tracking Accuracy Comparison

Results:

Method	Accuracy	Stability
A	High	Low
B	Moderate	High

Answer:

For long-term tracking, stability is critical to avoid drift or track loss accumulating over time. Method A's low stability makes it prone to failure over extended tracking despite higher instantaneous accuracy.

Conclusion: Method B is more suitable for long-term tracking due to its high stability; Method A is better suited to short-duration, high-precision tasks.

20. Surveillance Model Selection

Results:

Model	Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Answer:

Model A is fastest (30 ms) but its lower accuracy (82%) may be inadequate for critical surveillance; Model C offers the highest accuracy but its 120 ms latency clearly violates strict latency constraints.

Conclusion: if the latency threshold is strict (well under 50 ms), Model A is the only feasible option despite the accuracy trade-off; if the system can tolerate up to roughly 70 ms, Model B offers a much better accuracy-latency balance (90% at 70 ms) and is the recommended choice.

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Answer:

Accuracy drops sharply as noise increases (92% → 80% → 60%), showing the system lacks robustness to noisy input conditions.

Conclusion: recommend noise-reduction preprocessing (denoising filters, contrast/illumination normalization) and noise-robust training (data augmentation with synthetic noise) to improve high-noise performance.

22. Depth Estimation Error Analysis

Results:

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

Answer:

Method A has the lowest error (0.5 m), making it the most accurate for real-world depth-sensitive applications such as robotic grasping or AR placement; Method B is the least accurate.

Conclusion: recommend Method A when precision is critical; if Method A is infeasible due to hardware or cost, Method C (0.8 m) is the next-best alternative.

23. Multi-Model Comparison

Results:

Model	Accuracy	Speed	Robustness
A	88%	Fast	Low
B	92%	Moderate	High
C	95%	Slow	High

Answer:

Model C has the highest accuracy and high robustness but is slow, making it best for offline/critical, non-time-sensitive applications. Model B combines strong accuracy (92%) with high robustness at moderate speed, giving the best practical balance when both speed and robustness matter together.

Conclusion: recommend Model B as the best overall balanced model for most practical deployments; recommend Model C when speed is not a constraint and maximum accuracy/robustness is required.

24. Motion Detection under Crowd Density

Results:

Method	Sparse Crowd	Dense Crowd
Optical Flow	High	Low
Motion Models	Moderate	High

Answer:

Optical Flow performs best in sparse crowds but degrades in dense crowds, while Motion Models remain strong specifically in dense conditions despite being only moderate in sparse ones.

Conclusion: for systems that must operate reliably across varying crowd densities (e.g., public space monitoring), Motion Models are the more suitable overall choice, or a hybrid approach that switches between Optical Flow and Motion Models based on real-time density estimation.

25. Model Efficiency vs Accuracy

Results:

Model	Accuracy	Computation
A	90%	High

B	85%	Low
C	88%	Moderate

Answer:

For edge devices with limited compute/power, Model B is most efficient but sacrifices the most accuracy; Model C offers a good balance of accuracy (88%) and moderate computation.

Conclusion: recommend Model C for edge deployment as the best trade-off; recommend Model B only when compute/power is extremely constrained and some accuracy loss is acceptable.

26. Feature Detection under Illumination Changes

Results:

Method	Normal	Low Light
A	90%	60%
B	85%	75%

Answer:

Method A degrades severely in low light (a 30-point drop) compared to Method B's modest 10-point drop.

Conclusion: for applications with variable or low lighting (e.g., outdoor or 24-hour deployments), Method B is more robust despite its slightly lower normal-light accuracy, making it the recommended choice.

27. Video Processing Pipeline Comparison

Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Answer:

Pipeline A is best for applications prioritizing throughput and real-time processing; Pipeline C is best for offline, high-accuracy analytics where processing time is not constrained; Pipeline B is a near-real-time middle ground.

Conclusion: choose Pipeline A for real-time use, Pipeline C for batch/offline analysis, and Pipeline B when a balance of accuracy and reasonable speed is required.

28. Motion Model Evaluation

Results:

Model	Realism	Speed
A	High	Low
B	Moderate	High

Answer:

Model A is best suited to applications requiring high visual fidelity, such as film or offline simulation, despite its slow speed.

Model B is best suited to interactive or real-time applications like games or live previews, where speed matters more than perfect realism.

Conclusion: select Model A for fidelity-critical offline work and Model B for real-time interactive applications.

29. Detection under Occlusion

Results:

Method	Accuracy
A	70%
B	85%
C	90%

Answer:

Method C is the most robust to occlusion (90%), followed by B (85%); Method A (70%) is the weakest performer under occlusion.

Conclusion: recommend Method C for occlusion-heavy environments such as crowded scenes or industrial settings with overlapping objects; Method A is unsuitable where occlusion is frequent.

30. System Performance under Constraints

Results:

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Answer:

Model A offers the lowest latency and cost but the lowest accuracy; Model B balances higher accuracy (92%) with moderate latency and cost; Model C has the best accuracy but at the highest latency and cost, suiting only offline or high-budget precision deployments.

Conclusion: for most general real-time, budget-conscious deployments, Model B provides the best overall trade-off across accuracy, latency, and cost; Model A is preferred where cost/latency are hard constraints, and Model C where maximum accuracy is the priority regardless of cost.

Code of Conduct:

I V M SAI BHARGAV (192324126) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V M SAI BHARGAV